

Dakshi Goel



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EDUCATION

Bachelor of Technology, Indian Institute of Technology (Jammu, India)

July 2020 – May 2024

Majors - Electrical Engineering | Minors - Computer Science

GPA - 8.53/10.0

SKILLS

Programming Skills: Python, C++, SQL, NodeJS, ReactJS

Relevant Coursework: Machine Learning, Deep Learning, Data Structures and Algorithms, Database Management System, Computer Programming, Calculus, Linear Algebra, and Differential Equations.

Libraries/Platforms: NLTK, LangChain, Llama-Index, OpenAI, Gemini, Claude, FastAPI, REST API, BeautifulSoup, AWS, Postman, Pandas, Numpy, Matplotlib, Sklearn, TensorFlow, Keras, PyTorch.

PUBLICATIONS

Illumination Aware Multi-scale Attention Fusion Model for Underwater Image Enhancement

Published in *IEEE Transactions on Artificial Intelligence* ([Link](#)) | Indian patent filed ([No.202411053490](#))

- Designed an encoder-decoder network with attention-based fusion to capture and restore features at three different scales for underwater images, outperforming SOTA methods by 1.18 dB PSNR and 0.948 in SSIM.
- Proposed a multi-scale + LAB/LCH color loss function, enhancing color and texture fidelity, achieving UIQM 4.284.

Remote sensing data processing using convolutional neural networks for mapping alteration zones

2024 *International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS)* ([Link](#))

- Addressed class imbalance by combining SMOTE and GANs, achieving a 2-3% improvement in F1 scores for classification.
- Trained CNN and MLP for predicting alteration zones using hyperspectral remote sensing data, achieving an accuracy of 98.4%, beating the benchmark ML models.

WORK EXPERIENCE

Real Brokerage, AI Product Associate (Remote, Miami)

Dec 2024 - Aug 2024

- Analyzed 10,000+ chatbot conversations through Langfuse to resolve intent routing failures, using pattern matching (regex) and vector search. Implemented structured Q-tags to enrich chunks. Increased retrieval and routing accuracy by 20%.
- Enhanced LLM generation by designing few-shot, table-structured prompts and refining API trigger logic with context-aware reranking, reducing null/incorrect responses by 25% while improving end-to-end chatbot reliability.

Mozrest, Junior AI Developer (Remote)

Dec 2023 - Dec 2024

- Designed and deployed a GPT-3.5 Turbo for a task-oriented dialogue system with LangGraph on WhatsApp, integrating Stripe, and built a review response generation tool using few-shot prompting with sentiment control using GPT.
- Built and deployed forecasting models (ARIMA + LSTM) for restaurant sales and slot availability, achieving RMSE 7.2 and 93% accuracy while reducing latency by 125 ms through optimized 3,000-feature pipelines on AWS Lambda.

Infineon, AI, and Innovation Intern (Bangalore, India)

July 2023 - Dec 2023

- Developed RAG pipeline for semiconductor datasheets using Llama 13B model and FAISS.
- Achieved a BLEU score of 0.65. Deployed using FastAPI, integrating it into Infineon's no-code/low-code platform.

LTI Mindtree, Project Intern (Bangalore, India)

May 2023 - July 2023

- Implemented RAG pipeline and fine-tuned Llama2 13B model for code generation for Snowpark.
- Reduced hallucination rates by 40%. Developed a Streamlit chatbot interface and integrated it with FastAPI.

ML Course Mentor, IIT Jammu

- Mentored 50+ students across batches, monitored model implementation, code optimization, and debugging pipelines.
- Designed 6+ assignments, prepared lecture materials, and led 3+ research paper discussions in the class.

PROJECTS

Flipkart Grid Challenge

- Developed text-to-SQL LLM and RAG pipeline to interact with user metadata and fetch fashion trends in Postgres DB.
- Fine-tuned LLaMA-2 on custom-generated data for recommendations and utilized Redis Vector Store for low-latency.

Age and Gender Prediction from surveillance Video, Inter-IIT TechMeet, IIT KGP

Dec 2021

- Deployed ESRGANs to detect age and gender from low-quality surveillance video feed.
- Used RetinaFace, and a weighted regression cum classification model for face detection, increasing the F1 score by 0.1.

Audio Quality Assessment, DL Course, IIT Jammu

- Developed a transformer-based model with Dual Encoder Cross Attention using MFCC, MelSpectrogram, and Chroma CQT features. Optimized a 4-layer encoder-decoder with 4 attention heads for 1-5 scale audio quality prediction.
- Achieved PLCC: 0.828, SRCC: 0.823, KRCC: 0.629, surpassing SOTA benchmarks.

ACHIEVEMENTS

- Received **Smt. Godavari Devi Award** by IIT Jammu for the highest CGPA in Electrical Engineering.🏆
- **2nd place** in Flipkart Grid Robotics Challenge (Top 0.1% of 9,000 teams nationally) 🏆
- **2nd prize** in SEED Pitch Competition among 20 regional teams in Jammu and Kashmir .🏆
- Ranked **first** in Inter-IIT Tech Meet 10.0-Bosch Age and Gender Detection out of the 23 top IITs
- **Semi-finalist** in Google Girls Hackathon 2023 (Top 2.5% nationwide).
- **Top 5** among 75 teams pan India in the SERB-INAE Hackathon organized at Jadavpur University.

LEADERSHIP

- Student Head Representative, Career Development Services** 🏆 *Oct 2023-June,2024*
- Led a team of 70+, sourced 350+ job opportunities, organized 30+ workshops, launched a CEO Leadership Talk series.
- Dance Club Coordinator, IIT Jammu** 🏆 *July 2021 - Feb 2022*
- Managed a 50+ member team, organized 11+ events, and increased outreach by 80% through collaborations.